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LOW VISION

The relationship between visual function, duration and main causes of vision loss and falls in older people with low vision

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Abstract
Background Falls are an alarming health problem and a major cause of injury among the elderly. The healthcare cost associated with falls is considerable. Visual acuity has been found to be an independent risk factor for falls; however, the results are not unanimous. Moreover, other aspects of visual function such as visual field, contrast sensitivity and depth perception have not been adequately investigated in relation to falls. The aim of this study, therefore, was to determine the relationship between visual function, duration, and main causes of visual impairment, and falls in individuals with low vision.
Methods This was a cross-sectional study involving participants attending a public tertiary eye care hospital. Participants were mobile, aged 60 years or above, and had low vision (visual acuity >0.3 LogMAR in the better eye). Details about falls in the previous 12 months and other information were collected, and patients completed a questionnaire about activities of daily living. The duration and main causes of visual impairment, visual acuity, contrast sensitivity, depth perception, and visual field were assessed. Descriptive statistical analyses were performed to characterize the participants' sociodemographic and clinical data.
Results One hundred and twenty seven patients (53%; 67 males) with a mean age of 76.3±8.3 years were recruited. Thirty seven percent of the participants (n=47) had mild, 50% (n=64) moderate and 13% (n=16) severe visual impairment (>0.3–0.5; >0.5–1.0; and >1.0 LogMAR respectively). The frequencies of single and multiple falls were 42.5% and 12.6% respectively. Visual acuity, contrast sensitivity, depth perception, visual field, main cause, and duration of visual impairment were not significantly associated with falls (p>0.05). In multiple regression analyses, physical inactivity remained the only variable independently associated with falls in all models except for visual field. Overall, visually impaired people were three times more likely to fall if they were physically inactive.
Conclusions Visual function, duration and main causes of visual impairment are not independently associated with falls in people with low vision. However, a significant relationship between non-participation in physical activity and falls was found. Further work is required to investigate the association between vision-related factors and falls in older people with visual impairment.

Keywords Low vision · Falls · Visual function · Older individuals

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Introduction

Falls are an important health problem, as more than a third of community-living elderly individuals experience at least one fall annually [1–4]. Falls are the leading cause of injury-related deaths among elderly individuals [5, 6], and can result in restriction of mobility and activity, feelings of helplessness, loss of confidence, depression, and institutionalization [7]. Economically, the health care costs associated with falls in Australia were estimated at AU $500 million in 2001, and this figure is expected to almost triple by 2051 due to the anticipated ageing of the Australian population [8]. In the US, the cost of fatal and non-fatal falls is estimated to be even higher, standing at more than USD $19 billion (AUTHOR please correct this figure!) in 2000 [9]. Effective screening, prevention, and rehabilitation programs are required to prevent falls, as well the identification of the risk factors for falls. Falls have a multi-factorial etiology, and a considerable number of risk factors have been described as either personal or environmental [10]. Several population-based studies have identified reduced visual acuity as one of the most common risk factors for falls [11–13]. Also, individuals with compromised vision, compared to fully sighted populations, are almost twice as likely to fall, have recurrent falls, and sustain fractures [14]. However, other smaller prospective studies have failed to show any significant difference in visual acuity between fallers and non-fallers [15–20]. Other aspects of visual function are also not consistently related to falls. For example, a number of studies have demonstrated that contrast sensitivity can increase the risks of falls [15, 18, 20], while others have failed to replicate such findings [17, 21, 22]. Some studies have reported the role of depth perception and visual field in falls [23]. These findings indicate an equivocal role of visual function as a mechanism for falls, and suggest that further research in this area is needed.

In this study, we investigated evidence of the relationship between vision-related factors and falls in people with low vision. For example, we probed whether there was a relationship between severity of visual impairment and a higher frequency of falls. Such information is fundamental to prioritize prevention and rehabilitation programs for people who are at higher risk of falling within a visually-impaired population. Consequently, the aim of this study was to determine the relationship between visual function, severity, duration and causes of visual impairment, and falls in community-living elderly adults with low vision.

Materials and methods

All patients from the tertiary referral outpatient eye clinics at the Royal Victorian Eye and Ear Hospital (RVEEH) were initially approached between October 2005 and March 2006. They needed to be community-living individuals aged 60 years or above, have low vision (visual acuity>0.3 LogMAR (6/12 Snellen) in the better eye), be mobile and able to communicate in English or with a translator. All consenting participants provided information regarding demographics, general health, current medical history, present and past ocular history, presence of common risk factors for falls, use of medications, history of falls, participation in daily activities [24], level of physical activity, and education level. Ethics approval for this project was granted by the RVEEH Human Research Ethics Committee. This research adhered to the tenets of the Declaration of Helsinki.

Main outcome: falls

A fall was defined as an event which resulted in a person coming to rest unintentionally on the ground or other lower level [16]. All patients were asked to report the number of falls experienced during the previous month, 6 months and year. A patient was classified as a "Faller" if he or she reported a history of one or more falls in any of the above-mentioned periods. Otherwise, the patient was classified as a "Non-Faller". The specific characteristics of each fall were also recorded, namely the activity prior, location, presence and type of any obstacle involved, injuries, and prior warning signs [25].

Visual acuity

Visual acuity (VA) was tested individually and binocularly with a 4-m LogMAR chart. Patients with LogMAR>1.0 (6/60 Snellen) were allocated the following LogMAR equivalents: counting fingers (LogMAR=1.7); hand movements (LogMAR=2.0) and perception of light (LogMAR=2.3) [26].

Contrast sensitivity

The participants' contrast sensitivity was tested individually and binocularly using a 1-m Pelli–Robson contrast sensitivity chart [27].

Depth perception

Binocular depth perception was assessed using the Stereo Fly Test. Polarised vectograph plates are viewed through polarised viewers, allowing patients with stereopsis to view a 3-D image. If a loss in gross stereopsis is present, the patient will not be able to describe the figure of the housefly as solid or 3-D, it will appear flat. Patients were asked to use reading glasses if used habitually, and the polarised viewers were

worn over these. Patients were judged as having gross stereopsis or having a loss in gross stereopsis.

Visual field

Visual field loss was determined using the Humphrey Visual Field Analyser. The test used was the central 24-2 threshold test, using the Swedish interactive threshold algorithm (SITA) standard strategy on the Humphrey Visual Field Analyser (HFA) (Humphrey Zeiss Visual Field Analyser Model 750). As the name suggests, it can locate losses in the central 24 degrees of the visual field. Visual field loss was measured based on the number of abnormal stimulus locations, defined as those with statistical probability levels of $p<0.05$. A pattern deviation plot was used to determine the number of affected points [28].

Instrumental Activities of Daily Living (IADL)

The Lawton's Instrumental Activities of Daily Living scale [24] was also administered to ascertain the patients' level of functioning and independence. The scale rates the difficulty of performing eight daily living tasks, which are: 'using the telephone', 'getting to places beyond walking distance', 'grocery shopping', 'preparing meals', 'doing housework or handyman work', 'doing the laundry', 'taking medications', and 'managing money'. The response categories are 'unable to do', 'need some help', and 'need no help', and were coded as 0, 1 and 2 respectively. A summary score was generated where a participant needing no help with the eight items will produce a maximum score of 16. Conversely, a person unable to do any of the eight items will have a summary score of 0.

Statistical analysis

Descriptive statistical analyses were performed to characterize the participants' sociodemographic and clinical data. The distribution of variables such as visual acuity (logMAR), contrast sensitivity (db), visual field (points missed), and the IADL summary score was initially assessed for normality using the Kolmogorov–Smirnov, skewness and kurtosis test statistics, and found to be normally distributed. Statistical analyses using the χ^2 statistic for categorical data and Student's *t*-test or one-way ANOVA for continuous data were performed to identify factors that differed between fallers and non-fallers. Binary logistic regression analyses for predictors of falls were performed for variables which were significant in the univariate analyses. Data were analyzed using the SPSS statistical software (Version 14.0.1, SPSS Science, Chicago, IL, USA). A p value<0.05 was used to indicate statistical significance.

Results

One hundred and twenty-seven patients (53%; 67 males) with a mean age of 76.3±8.3 years were recruited for the study (Table 1). Only 39.4% of the participants ($n=50$) were born in Australia. Ninety-five participants (74.8%) were involved in regular physical activity. More than 60% of the participants ($n=75$) rated their health as being 'good', and the most common medical conditions were diabetes and arthritis which affected 35% ($n=44$) and 38% ($n=48$) of the participants respectively. The mean IADL summary score was 11.6±4.3 (range 0–16), which suggests that overall the participants were generally functional and independent.

The participants' clinical characteristics and visual functioning are shown in Table 2. The most common eye conditions were glaucoma (33.1%, $n=42$) and cataract in either eye (21.3%, $n=27$). The mean ± SD for best corrected binocular visual acuity (BVCA) and contrast sensitivity were 0.72±0.47 LogMAR (6/30) and 1.02±0.47 db respectively. Over forty percent of our participants ($n=53$) had a lack of gross stereopsis, and the mean ± SD points missed on the visual field test was 16.03±12.6. Overall, 37% of the participants ($n=47$) had mild, 50% ($n=64$) moderate and 13% ($n=16$) severe visual impairment (VA (better eye) >0.3–0.5; >0.5–1.0; and >1.0 LogMAR respectively).

Frequency and characteristics of falls

The frequency of recalled falls was 42.5% with 54 patients reporting 111 falls in the last 12 months. However, the participants could only recall the details of 71 of these falls, suggesting a recall bias of 36%. Of these, 16 (29.6%) reported more than one fall. Overall, 37 patients (29.1%) reported having fallen at least once in the previous 6 months, while 18 (14.2%) reported having fallen at least once in the last month.

Almost 65% of the falls ($n=46$) happened indoors (e.g., the bedroom, bathroom). The majority of falls (60%) occurred at the patients' home and surrounding environments (garden or backyard areas). Negotiating obstacles (e.g., curbs and steps) and slippery surfaces contributed to more 50% of all falls. 'Walking' was the most common activity in which the participants were involved at the time of the fall (62%). Only 18.3% of the falls were preceded by any warning sign such as dizziness, palpitations, or weakness. More than one third of the falls resulted in bruises and grazes, 15% in cuts and almost 10% in fractures. Approximately 40% of the patients were unable to get up by themselves after the fall, and 34.4% sought medical attention following a fall.

Table 1 Characteristics of the 127 visually impaired participants of the study

Patients' characteristics		Total (127)	Non-faller (n=73)	Faller (n=54)	P value
Age (years)	Mean ± SD	76.3±8.3	76.2±8.4	76.4±8.7	0.88
Gender	Female	60 (47.2%)	33 (55%)	27 (45%)	0.72
Country of birth	Australia	50 (39.7%)	25 (50%)	25 (50%)	0.21
Smoker	Never	53 (41.7%)	27 (50.9%)	26 (49.1%)	0.36
	Current	13 (10.3%)	7 (53.8%)	6 (46.2%)	
	Past	61 (48.0%)	39 (63.9%)	22 (36.1%)	
Other conditions	Diabetes	48 (37.8%)	28 (38.4%)	20 (37.0%)	0.81
	Arthritis	44 (37.8%)	22 (30.1%)	22 (40.7%)	0.26
	History of stroke	8 (6.3%)	1 (1.4%)	7 (13.0%)	0.01[b]
	Parkinson's disease	1 (0.8%)	0 (0.0)%	1 (100%)	0.43
	COPD	1 (0.8%)	0 (0.0)%	1 (100%)	0.43
	Urinary incontinence	35 (27.6%)	18 (51.4%)	17(48.6%)	0.43
	Hearing impairment	52 (40.9%)	24 (46.2%)	28 (53.8%)	0.05
	Foot problems	40 (31.5%)	21 (52.5%)	19 (47.5%)	0.45
Use of	Anti-depressants	8 (6.3%)	3 (37.5%)	5 (62.5%)	0.28
	Assistive device	38 (29.9%)	16 (42.1%)	22 (57.9%)	0.18[b]
Physical activity	Yes	95 (74.8%)	62 (65.3%)	33 (34.7%)	<0.01[b]
	No	32 (25.2%)	11 (34.4%)	21 (65.6%)	
General health	Very good/excellent	29 (22.8%)	19 (65.5%)	10 (34.5%)	0.85
	Good	46 (36.2%)	28 (60.9%)	18 (31.1%)	
	Poor/fair	52 (40.9%)	26 (50%)	26 (50%)	
Alcohol	Regular consumption	47 (37.0%)	30 (63.8%)	17 (36.2%)	0.35
IADL[a]	Summary score (mean±SD)	11.6±4.3	12.8±3.4	10.1±4.1	<0.01[b]

[a] IADL: Instrumental Activities of Daily Living summary score ranging between 0 and 16, where a low score indicates substantial difficulty undertaking activities of daily living.

[b] $p<0.05$

Table 2 Clinical characteristics of the 127 visually impaired participants of the study

Characteristics		Total (n=127)	Non-faller (n=73)	Faller (n=54)	P value[b]
Eye conditions	AMD	12 (9.4%)	9 (75%)	3 (25%)	0.11
	Cataract	27 (21.3%)	17 (63%)	10 (27%)	
	DR	25 (19.7%)	12 (48%)	13 (52%)	
	Glaucoma	42 (33.1%)	27 (64.3%)	15 (35.7%)	
	Other	21 (16.5%)	8 (38.1%)	13 (61.9%)	
Depth perception	Gross stereopsis[a]	72 (56.7%)	42 (58.3%)	30 (41.7%)	0.85
Visual acuity (LogMAR-mean ± SD)	Better eye	0.73±0.49	0.73±0.49	0.73±0.48	0.96
	Worse eye	1.59±0.65	1.56±0.65	1.64±0.64	0.50
	Binocular	0.72±0.47	0.72±0.49	0.71±0.46	0.92
Contrast sensitivity (db-mean ± SD)	Worse eye	1.04±0.45	0.49±0.53	0.39±0.48	0.33
	Better eye	0.45±0.51	1.10±0.42	0.96±0.47	0.12
	Binocular	1.02±0.47	1.10±0.45	0.93±0.50	0.11
Visual field (mean ± SD)	Points missed	16.03±12.6	13.75±14.3	17.27±10.3	0.28
Duration of impairment (years)	Median (range)	4 (1–65)	4.0 (1–65)	4.5 (1–57)	0.25

AMD = age-related macular degeneration, DR = diabetic retinopathy

[a] Data missing for two patients

[b] p values derived from chi-squared and ANOVA tests, as appropriate.

Comparison between fallers and non-fallers

In the univariate analyses, fallers compared to non-fallers were significantly less likely to be involved in vigorous physical activity (χ^2; $p<0.05$) (Table 1). Overall, they also experienced significantly greater difficulty performing activities of daily living (ANOVA; $p<0.05$). A significantly greater proportion of fallers reported a history of stroke and making use of an assistive device while walking (χ^2; $p<0.05$). There were no differences between fallers and non-fallers for eye conditions and the duration of vision impairment (χ^2 and *t*-test; $p>0.05$, Table 2). Critically, our analyses found no statistically significant differences for visual acuity, depth perception, contrast sensitivity and visual field between fallers and non fallers (ANOVA; $p>0.05$; Table 2).

Multivariate analysis

Although visual acuity, contrast sensitivity, depth perception, and visual field were not found to be significantly associated with falls, each was included separately in a regression model together with age, gender, and other variables significantly related with falls. Non-participation in physical activity was found to be independently significantly associated with recalled falls in three of the four models (Table 3).

Overall, a strong relationship was observed between falling, visual impairment and non-involvement in any form of physical activity.

Discussion

Our results indicate that more than 40% of individuals with low vision attending a public tertiary eye care hospital fell in the past 12 months. The frequency of falls reported here is similar to or even higher (28–35%) than that in normally sighted community-dwelling elderly individuals [1, 2, 4, 16, 19], suggesting that visually impaired people are at a substantial risk of falls. Furthermore, almost 30% of our

Table 3 Four binary logistic regression models showing the relationship between falls in the last 12 months and visual impairment

	Variables entered	Coefficient	*P* value	Odds ratio	95.0% confidence interval for odds ratio	
					Lower	Upper
Model 1	Visual acuity	-0.09	0.82	0.91	0.40	2.04
	Gender	-0.01	0.98	0.98	0.46	2.13
	Age	-0.03	0.19	0.96	0.92	1.02
	Physical Activity[a]	**1.18**	**0.01**	**3.27**	**1.34**	**7.96**
	History of Stroke	0.91	0.12	2.48	0.77	7.98
	Summary IADL score	-0.27	0.24	0.76	0.47	1.21
Model 2	Contrast sensitivity	-0.59	0.17	0.55	0.23	1.29
	Gender	0.00	0.98	1.01	0.46	2.21
	Age	-0.03	0.20	0.96	0.91	1.02
	Physical Activity[a]	**1.17**	**0.01**	**3.25**	**1.31**	**8.05**
	History of Stroke	0.85	0.17	2.36	0.68	8.19
	Summary IADL score	-0.24	0.2	0.78	0.49	1.24
Model 3	Depth perception	0.36	0.37	1.44	0.64	3.25
	Gender	0.04	0.91	1.05	0.48	2.26
	Age	-0.02	0.36	0.97	0.92	1.03
	Physical Activity[a]	**1.23**	**0.01**	**3.42**	**1.40**	**8.34**
	History of Stroke	0.73	0.24	2.08	0.61	7.03
	Summary IADL score	-0.29	0.21	0.74	0.46	1.18
Model 4	Visual field	-0.04	0.06	0.96	0.92	1.00
	Gender	-0.23	0.64	0.79	0.29	2.11
	Age	-0.04	0.25	0.96	0.89	1.03
	Physical Activity	1.08	0.08	2.96	0.87	10.11
	History of Stroke	0.52	0.53	1.68	0.32	8.96
	Summary IADL score	-0.23	0.41	0.79	0.46	1.38

[a] denotes a significant ($p<0.05$) independent association with having a fall in each model.

participants had had more than one fall in the last 12 months. This finding is lower than that reported in a recent prospective study of normally sighted community-dwelling elderly people in which 50% of single fallers reported multiple falls [15]. The difference is likely to be due to reliance on memory recall within retrospective studies, resulting in an underestimate of falls [29].

Only non-participation in physical activity was found to be independently and significantly linked with falls. Our finding is not unusual, as several studies using other population samples have also shown inactivity to be a consistent risk factor for falls and moving from being sedentary to moderately active can reduce the risks of falls and hip fractures by 20–40% [16, 30, 31]. Physical activity reflects aspects of mobility, dynamic stability and musculoskeletal system function. Previous investigations that have shown measures of muscle weakness, such as the inability to get up from a chair without using the hands, to predict falls [17, 19]. Given that these factors are readily modifiable with appropriate intervention strategies, further work is required to examine these elements and their associations with falls within the visually impaired population.

The four aspects of visual function were not found to be associated with falls. Several population-based studies have found visual acuity (<6/12) to be significantly associated with falls [13, 21, 32], including injurious falls [32, 33]. Whilst visual impairment (absence or presence) may be an independent predictor of falls in a heterogeneous population, our results indicate that the degree or level of visual acuity has limited association with falls in this specific population. It has been argued that visual acuity plays a more significant role as a falls risk factor when the participants are dependent [33]. It should be noted that our sample were mobile and independent community-dwelling individuals, and therefore the role of visual acuity as an independent predictor of falls may be less important.

Furthermore, over 90% of our participants had mild or moderate vision loss, and it is possible that a floor effect in the range of visual impairment may help to explain our lack of association between level of visual acuity and falls. Future studies, using a larger sample size and an even representation of visual impairment, are needed to confirm these hypotheses and to determine more precisely the level of visual acuity at which the risk of falls increases.

That visual field loss was not found to be associated with falls is perhaps also not surprising, as there is a lack of consensus on this issue. Several studies have shown an increased risk of falls in patients with compromised visual field [13, 34, 35], while others have shown no association [22, 36]. Similarly, some studies have found contrast sensitivity to predict falls [37], while others have found no significant association [21, 22]. Equally, the association of depth perception with falls has been ambiguous, as some studies have demonstrated a link [15], while others have found none [22, 35]. Our finding that there is no association between falls and eye conditions is also not unprecedented, as there has been conflicting evidence on the role of eye diseases as potential risk factors for falls [11, 12, 32, 38].

In conclusion, this study found that more than two out of five individuals with low vision fall annually, suggesting that a substantial proportion of this specific population is vulnerable to falling and its debilitating impact. Four aspects of visual function were not found to be associated with falling, suggesting that once the visual system deteriorates, other factors such as participation in physical activity may be more significant. Such information is valuable in the planning of prevention and rehabilitation programs for people who are at higher risk of falling within a visually-impaired population, and could be the focus of larger studies in the future. Due to the cross-sectional nature of this study and the relatively small sample size, further investigation is required to determine whether a fear of falling or a previous fall limits participation in physical activity and, additionally, that the non-involvement in physical activity increases the risk of falling. Future prospective research involving a larger sample from diverse settings and including formal assessment of functional gait, mobility, balance, and muscular strength is also required to identify the independent predictors of falls in individuals with low vision.

References

1. Campbell AJ, Reinken J, Allan BC, Martinez GS (1981) Falls in old age: a study of frequency and related clinical factors. Age Ageing 10:264–270
2. Prudham D, Evans JG (1981) Factors associated with falls in the elderly: a community study. Age Ageing 10:141–146
3. Tinetti ME, Speechley M, Ginter SF (1988) Risk factors for falls among elderly persons living in the community. New England Journal of Medicine 319:1701–1707
4. Blake AJ, Morgan K, Bendall MJ, Dallosso H, Ebrahim SB, Arie TH, Fentem PH, Bassey EJ (1988) Falls by elderly people at home: prevalence and associated factors. Age Ageing 17:365–372
5. Dunn C, Sadkowsky K, Jelfs P (2002) Trends in Deaths: Australian data, 1987–1998 with updates to 2000. Australian Institute of Health and Welfare, Canberra
6. Baker SP, Harvey AH (1985) Fall injuries in the elderly. Clin Geriatr Med 1:501–512
7. Tinetti ME, Powell L (1993) Fear of falling and low self-efficacy: a case of dependence in elderly persons. J Gerontol 48 Spec No:35–38
8. Moller J (2003) Projected costs of fall related injury to older persons due to demographic changes in Australia. Commonwealth Department of Health and Ageing
9. Stevens JA, Corso PS, Finkelstein EA, Miller TR (2006) The costs of fatal and non-fatal falls among older adults. Inj Prev 12:290–295, doi:10.1136/ip. 2005.011015

10. Connell BR (1996) Role of the environment in falls prevention. Clin Geriatr Med 12:859–880
11. Ivers R, Cumming R, Mitchell P (2002) Poor vision and risk of falls and fractures in older Australians: the Blue Mountains Eye Study. New South Wales Public Health Bulletin 13:8–10
12. Coleman AL, Stone K, Ewing SK, Nevitt M, Cummings S, Cauley JA, Ensrud KE, Harris EL, Hochberg MC, Mangione CM (2004) Higher risk of multiple falls among elderly women who lose visual acuity. Ophthalmology 111:857–862
13. Ivers RQ, Cumming RG, Mitchell P, Attebo K (1998) Visual impairment and falls in older adults: the Blue Mountains Eye Study. J Am Geriatr Soc 46:58–64
14. Legood R, Scuffham P, Cryer C (2002) Are we blind to injuries in the visually impaired? A review of the literature. Inj Prev 8:155–160
15. Lord SR, Dayhew J (2001) Visual risk factors for falls in older people. J Am Geriatr Soc 49:508–515
16. Tinetti ME, Speechley M, Ginter SF (1988) Risk factors for falls among elderly persons living in the community. N Engl J Med 319:1701–1707
17. Nevitt MC, Cummings SR, Kidd S, Black D (1989) Risk factors for recurrent nonsyncopal falls. A prospective study. JAMA 261:2663–2668
18. Lord SR, Clark RD, Webster IW (1991) Visual acuity and contrast sensitivity in relation to falls in an elderly population. Age Ageing 20:175–181
19. Campbell AJ, Borrie MJ, Spears GF (1989) Risk factors for falls in a community-based prospective study of people 70 years and older. J Gerontol 44:M112–M117
20. de Boer MR, Pluijm SM, Lips P, Moll AC, Volker-Dieben HJ, Deeg DJ, van Rens GH (2004) Different aspects of visual impairment as risk factors for falls and fractures in older men and women. J Bone Miner Res 19:1539–1547
21. Klein BE, Klein R, Lee KE, Cruickshanks KJ (1998) Performance-based and self-assessed measures of visual function as related to history of falls, hip fractures, and measured gait time. The Beaver Dam Eye Study. Ophthalmology 105:160–164
22. Friedman SM, Munoz B, West SK, Rubin GS, Fried LP (2002) Falls and fear of falling: which comes first? A longitudinal prediction model suggests strategies for primary and secondary prevention. J Am Geriatr Soc 50:1329–1335
23. Black A, Wood J (2005) Vision and falls. Clin Exp Optom 88:212–222
24. Lawton MP, Brody EM (1969) Assessment of older people: self-maintaining and instrumental activities of daily living. Gerontologist 9:179–186
25. Bloem BR, Grimbergen YA, Cramer M, Willemsen M, Zwinderman AH (2001) Prospective assessment of falls in Parkinson's disease. J Neurol 248:950–958
26. Nicholas S, Riley A, Patel H, Neveldson B, Purdie G, Wells AP (2006) Correlations between optical coherence tomography measurement of macular thickness and visual acuity after cataract extraction. Clinical and Experimental Ophthalmology 34:124–129
27. Pelli DG, Robson JG, Wilkins AJ (1988) The design of a new letter chart for measuring contrast sensitivity. Clin Vis Sci 2:187–199
28. Hudson C, Flanagan JG, Turner GS, Chen HC, Young LB, McLeod D (1998) Influence of laser photocoagulation for clinically significant diabetic macular oedema (DMO) on short-wavelength and conventional automated perimetry. Diabetologia 41:1283–1292
29. Cummings SR, Nevitt MC, Kidd S (1988) Forgetting falls. The limited accuracy of recall of falls in the elderly. J Am Geriatr Soc 36:613–616
30. Gillespie LD, Gillespie WJ, Robertson MC, Lamb SE, Cumming RG, Rowe BH (2001) Interventions for preventing falls in elderly people. Cochrane Database Syst Rev: CD000340
31. Gregg EW, Pereira MA, Caspersen CJ (2000) Physical activity, falls, and fractures among older adults: a review of the epidemiologic evidence. J Am Geriatr Soc 48:883–893
32. Felson DT, Anderson JJ, Hannan MT, Milton RC, Wilson PW, Kiel DP (1989) Impaired vision and hip fracture. The Framingham Study. J Am Geriatr Soc 37:495–500
33. Koski K, Luukinen H, Laippala P, Kivela SL (1998) Risk factors for major injurious falls among the home-dwelling elderly by functional abilities. A prospective population-based study. Gerontology 44:232–238
34. Ramrattan RS, Wolfs RC, Panda-Jonas S, Jonas JB, Bakker D, Pols HA, Hofman A, de Jong PT (2001) Prevalence and causes of visual field loss in the elderly and associations with impairment in daily functioning: the Rotterdam Study. Arch Ophthalmol 119: 1788–1794
35. Klein BE, Moss SE, Klein R, Lee KE, Cruickshanks KJ (2003) Associations of visual function with physical outcomes and limitations 5 years later in an older population: the Beaver Dam Eye Study. Ophthalmology 110:644–650
36. Glynn RJ, Seddon JM, Krug JH Jr, Sahagian CR, Chiavelli ME, Campion EW (1991) Falls in elderly patients with glaucoma. Arch Ophthalmol 109:205–210
37. Klein BE, Klein R, Knudtson MD, Lee KE (2003) Relationship of measures of frailty to visual function: the Beaver Dam Eye Study. Trans Am Ophthalmol Soc 101:191–196 discussion 196–199
38. Ivers RQ, Cumming RG, Mitchell P, Simpson JM, Peduto AJ (2003) Visual risk factors for hip fracture in older people. J Am Geriatr Soc 51:356–363